

Contact Information

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Summary

M.Sc. Wind Energy Engineering student at Hochschule Flensburg, with hands-on experience in wind farm planning, energy yield assessment, noise and shadow flicker analysis and bat curtailment strategies. Backed by a strong foundation in aeroelastic simulation, structural dynamics and Python automation, and four peer-reviewed publications from control systems research. Experienced in technical documentation and project coordination across industrial engineering environments.

Education

03/2023 – present

M.Sc. Wind Energy Engineering, (current GPA: 1.9)

Hochschule Flensburg, Flensburg, Germany

- Performed wind farm energy yield assessments using WindPro and QGIS, including noise calculation, shadow flicker analysis, site suitability assessment and bat curtailment strategies in compliance with environmental regulations.
- Conducted aeroelastic load simulations for a 20 MW offshore monopile wind turbine using OpenFAST, Python and MATLAB, covering IEC 61400-1 Design Load Cases.
- Developed Python scripts to automate OpenFAST simulation workflows, reducing manual pre- and post-processing effort by **approximately 80%**.

09/2014 – 01/2017

M.Sc. Mechanical Engineering

Hakim Sabzevari University, Mashhad

Thesis: Developed and implemented a control system for an autonomous underwater welding robot enabling precision movement and positioning, using MATLAB/Simulink. Research resulted in four peer-reviewed publications.

09/2008 – 06/2013

B.Sc. Mechanical Engineering

Azad University, Tehran

Designed a high-frequency screening machine using CATIA V5 and performed structural and vibration analysis using ANSYS.

Work Experience

01/2022 – 02/2023

Junior Development Engineer – *Paydar, Tehran*

- Designed and optimized mechanical components for solar mounting systems using SolidWorks, with a focus on structural integrity and reliability.
- Performed structural analyses using ANSYS, optimizing existing designs and achieving a **10% reduction** in scrap rate.
- Developed and maintained **technical documentation** and engineering data within SAP-based systems.

03/2017 – 12/2021

Mechanical Engineer – *TehranRigzar, Tehran*

- Designed large mechanical assemblies for mining equipment, including high-frequency vibrating screens, using CATIA V5.
- Conducted dynamic and structural analyses using ANSYS to optimize designs, achieving a **20% improvement** in system performance and reduced material costs.
- Coordinated project teams and led progress meetings to ensure timely delivery of project objectives.

Key Technical Skills

- Wind Energy: OpenFast, WindPro, QGIS
- Programming & Data Analysis: Python, MATLAB, Power BI
- CAD: CATIA V5, SolidWorks, Siemens NX
- CAE: ANSYS Workbench, ABAQUS
- Project Management & Office: MS Project, Excel, PowerPoint, Word, Documentation

Certifications

- **Python** — HarvardX
- **ANSYS** — IEEE
- **SOLIDWORKS** — Iran Technical & Vocational Training Organization
- **CATIA V5** — AmirKabir University of Technology

Languages

- English – IELTS overall score 7
- German – Intermediate B1 (actively improving)